Week 2 lecture: convex sets and convex functions

A set is convex when the straight segment between any two of its points stays inside the set. Balls, half-spaces, and polyhedra are convex. A crescent shape is not.

A function f is convex when its epigraph is a convex set.
Equivalently, for every x, y and every t between 0 and 1:
f(t x + (1 - t) y) <= t f(x) + (1 - t) f(y).

Why convexity matters for optimisation

For a convex function every local minimum is a global minimum. That is why gradient descent with a small enough step size finds the best answer on a convex problem, and why it can get stuck on a non-convex one.

Second-order test: a twice-differentiable function is convex when its Hessian matrix is positive semidefinite everywhere.